15_Jensen's inequality

Jensen's Inequality (Deterministic Form)

Let f be a **convex** function. Then for all x, y and all $\lambda \in [0,1]$,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

Interpretation:
For a convex function, the function value at a weighted average of points is **less than or equal to** the weighted average of the function values.

Example 01: Convex Function

$$f(x) = x^2$$

Consider the function

$$f(x) = x^2$$

Since

$$f''(x) = 2 > 0,$$

the function is **convex**.

Step 1: Choose two points and a weight

Let

$$x = 1, y = 3, \lambda = 0.5$$

Step 2: Compute both sides of Jensen's inequality

Left-hand side:

$$\lambda x + (1-\lambda)y = 0.5(1) + 0.5(3) = 2$$
$$f(2) = 2^2 = 4$$

Right-hand side:

$$\lambda f(x) + (1-\lambda)f(y) = 0.5(1^2) + 0.5(3^2)$$
$$= 0.5(1) + 0.5(9) = 0.5 + 4.5 = 5$$

Step 3: Compare

$$4 \leq 5$$

The inequality holds.

Since

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y),$$

the function $f(x) = x^2$**satisfies Jensen's inequality**, confirming that it is a **convex function**.

Example 02: Non-Convex Function

$f(x) = -x^2$**(Concave Case)**

Consider the function

$$f(x) = -x^2$$

Since

$$f''(x) = -2 < 0,$$

the function is **concave**, not convex.

Let

$$x = 1, y = 3, \lambda = 0.5$$

Step 1: Compute the left-hand side

$$\lambda x + (1 - \lambda)y = 0.5(1) + 0.5(3) = 2$$
$$f(2) = -(2^2) = -4$$

Step 2: Compute the right-hand side

$$0.5f(1) + 0.5f(3) = 0.5(-1^2) + 0.5(-3^2)$$
$$= 0.5(-1) + 0.5(-9) = -0.5 - 4.5 = -5$$

Step 3: Compare both sides

$$-4 \nleq -5$$

The Jensen inequality for **convex** functions fails.

Since

$$f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y),$$

the function $f(x) = -x^2$ **does not satisfy Jensen's inequality in convex form**, confirming that it is **not convex**.

(Indeed, the inequality is reversed, as expected for a concave function.)